

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Third Semester of B. Pharm. Examination
University Theory Examination April/May 2016
PH219 Pharmaceutical Engineering-III

Date: 18/04/2016, Monday Time: 01:30 p.m. to 04:30 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

SECTION – I

- Q 1 Attempt all questions. Each question is of one mark. 20**
1. Which of the following is a mechanism of size reduction?
[A] Cutting
[B] Compression
[C] Attrition
[D] All of the above
 2. Which of the following is a factor affecting size reduction?
[A] Hardness
[B] Toughness
[C] Stickiness
[D] All of the above
 3. Roller mill works on a principle of
[A] Impact
[B] Attrition
[C] Cutting
[D] Compression
 4. Size reduction of vitamins and antibiotics will be carried out using
[A] Roller Mill
[B] Ball Mill
[C] Cutter Mill
[D] Fluid Energy Mill
 5. Ball mill works on a principle of
[A] Impact
[B] Attrition
[C] None
[D] Both
 6. Number of sieve indicating the number of meshes included in a length of
[A] 25.4 mm
[B] 23.5 mm
[C] 26.4 mm
[D] 22.7 mm

7. A powder sample passes through 10# sieve is called as
 - [A] Coarse
 - [B] Fine
 - [C] Very fine
 - [D] Medium
8. Which of the following is not a sedimentation method?
 - [A] Hammer Mill
 - [B] Elutriation
 - [C] Sedimentation Tank
 - [D] Cyclone Separator
9. Which of the following method is suitable for grinding of material having water content more than 50%
 - [A] Wet grinding
 - [B] Dry grinding
 - [C] Cutting
 - [D] Pressure Grinding
10. Most commonly used metal for making the sieve
 - [A] Stainless Steel
 - [B] Aluminium
 - [C] Zinc
 - [D] Tin
11. Which of the following is not a type of mixture?
 - [A] Positive mixture
 - [B] Negative mixture
 - [C] Neutral Mixture
 - [D] None
12. Mixing promotes
 - [A] physical change
 - [B] formation of dispersion
 - [C] reaction
 - [D] all of the above
13. Mixing process is affected by
 - [A] material density
 - [B] particle size
 - [C] particle shape
 - [D] all of above
14. Turbine mixer is a example of
 - [A] solid mixer
 - [B] gas mixer
 - [C] semisolid mixer
 - [D] liquid mixer
15. Which of the following is used for mixing of powder samples?
 - [A] Agitator mixers
 - [B] Tumbler mixers
 - [C] None
 - [D] Both
16. which of the following is not a type of mechanical injury
 - [A] crushing
 - [B] cutting
 - [C] tearing
 - [D] burning

17. Substance exist in more than one crystalline form is called as
 [A] Mumorphs
 [B] Decamorphs
 [C] Paramorphs
 [D] Polymorphs
18. Various stages in crystallization
 [A] Nucleation
 [B] Growth
 [C]Evapoartion
 [D] Both A and B
19. The crystal size distribution depends on
 [A]number of nuclie formed
 [B]relationship between nucleation and growth
 [C]Nucleation alone
 [D]Both A and B
20. which of the following is not a type of fire extinguisher
 [A] CO₂
 [B] Dry Chemical
 [C] Halon
 [D] O₂

SECTION – II

- Q 2** Attempt any **FOUR** of the following
- | | | |
|----------|---|-----------|
| A | Define size reduction. What are the objectives of size reduction? | 05 |
| B | Write note on Ball Mill. | 05 |
| C | Explain the mechanisms of size reduction in detail. | 05 |
| D | Write note on cyclone separator. | 05 |
| E | Expalin the pharmacopoeial standards of sieve. | 05 |
| F | Write note on chemical hazards. | 05 |

SECTION – III

- Q 3** Attempt any **FOUR** of the following
- | | | |
|----------|---|-----------|
| A | Explain principal, construction, working, advantage and disadvantages of Sigma-Blade Mixer | 05 |
| B | Enlist and Explain (only one) principal, construction, working, advantage and disadvantage of mixer for semisolid mixing. | 05 |
| C | Explain principal, construction, working, advantage and disadvantages of Silverson Emulsifier | 05 |
| D | Explain Mier'sSupersaturationTherory. | 05 |
| E | Explain in detail spherical crystallization and its application in pharmacy. | 05 |
| F | Explain crystallizer for thermolabile material: Principal, Construction, Working, advantages and disadvantages. | 05 |
- Q 4** Attempt any **FOUR** of the following
- | | | |
|----------|---|-----------|
| A | Explain in detail mechanism of crystallization. | 05 |
| B | Explain in detail principal, construction, working, advantages and disadvantages of Swenson Walker Crystallizer | 05 |
| C | What is crystal? Describe various form of crystals. | 05 |
| D | Differentiate Analog and Digital measurement system. | 05 |
| E | Explain principal, contstruction and working of thermocouple based temperature controlled device. | 05 |
| F | Explain Acid-Base Autotitrator. | 05 |